

**From Mutual-Information Laplacians to Cosmological Dynamics:
A First-Principles Derivation of the Effective Stress–Energy
Tensor**

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Abstract

We present a fully self-contained derivation of the effective macroscopic stress-energy tensor generated by a pre-geometric quantum substrate whose structure is encoded in the spectrum of a mutual-information (MI) Laplacian. Starting from a canonical partition function over the nonzero eigenmodes of the MI-Laplacian, we show that homogeneous spectral scaling under global spatial expansion uniquely determines the energy-density scaling, the equation-of-state dictionary $w_{\text{eff}} = \alpha/3 - 1$, and the perfect-fluid form of $T_{\mu\nu}^{\text{eff}}$. The derivation provides a microscopic foundation for emergent-geometry cosmology and identifies the de Sitter phase as a strict topological attractor of the information-saturated substrate. Under total information saturation, relational connectivity becomes distance-independent, forcing the entanglement decay exponent to $\alpha = 0$, which natively yields a true cosmological constant phase ($w_{\text{eff}} = -1$). The resulting mapping $T_{\mu\nu}^{\text{eff}}(\mathcal{L}_{\text{MI}})$ is universal, model-independent, and requires no assumptions beyond the spectral properties of the MI-Laplacian.

I. INTRODUCTION

A central challenge in emergent-gravity and quantum-information approaches to space-time is the construction of a rigorous bridge between microscopic relational data and macroscopic cosmological dynamics. While numerous models demonstrate how geometry may arise from entanglement patterns, far fewer provide an explicit derivation of the effective stress-energy tensor sourcing the Einstein equations. In this work we close this gap by deriving $T_{\mu\nu}^{\text{eff}}$ directly from the spectral properties of the mutual-information (MI) Laplacian of a pre-geometric quantum substrate.

The derivation is fully self-contained and independent of any specific microscopic Hamiltonian. It relies only on (i) the MI-Laplacian spectrum, (ii) a canonical partition function over its nonzero eigenmodes, and (iii) the homogeneous scaling of eigenvalues under global spatial expansion. These ingredients are sufficient to determine the energy density, pressure, and equation of state of the emergent cosmological fluid.

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II. RELATIONAL SUBSTRATE AND MI-LAPLACIAN

Consider a quantum many-body system partitioned into subregions labelled by indices i, j . The mutual information between two subregions is

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (1)$$

where $S(\rho)$ is the von Neumann entropy. The matrix $I(i, j)$ serves as a relational measure of proximity: highly entangled regions are “close” in the emergent geometry.

From the normalized mutual-information matrix we construct a symmetric adjacency matrix A_{ij} and the corresponding combinatorial Laplacian

$$L_{\text{MI}} = D - A, \quad (2)$$

where $D_{ii} = \sum_j A_{ij}$. Let the eigenvalues of L_{MI} be

$$0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{N-1}. \quad (3)$$

These eigenvalues encode the stiffness of relational degrees of freedom and constitute the fundamental microscopic spectrum of the substrate.

III. CANONICAL PARTITION FUNCTION

We define a canonical partition function over the nonzero eigenmodes:

$$Z(\beta) = \sum_{n=1}^{N-1} e^{-\beta \lambda_n}, \quad (4)$$

where β is an inverse entanglement temperature. The internal energy is

$$U(\beta) = -\frac{\partial}{\partial \beta} \ln Z(\beta) = \frac{\sum_{n=1}^{N-1} \lambda_n e^{-\beta \lambda_n}}{\sum_{n=1}^{N-1} e^{-\beta \lambda_n}}. \quad (5)$$

The effective macroscopic energy density is

$$\rho_{\text{eff}}(a) = \frac{U(\beta, a)}{V_{\text{eff}}(a)}, \quad (6)$$

where $V_{\text{eff}}(a)$ is the emergent geometric volume at scale factor a .

IV. HOMOGENEOUS SPECTRAL SCALING

A key structural property of the MI-Laplacian is its homogeneous scaling under global spatial expansion. If coordinate distances scale as $d(i, j) \rightarrow a d(i, j)$, then the Laplacian eigenvalues transform as

$$\lambda_n(a) = a^{-\alpha} \lambda_n, \quad (7)$$

where α is the entanglement decay exponent defined by $I(i, j) \sim d(i, j)^{-\alpha}$.

The scale-dependent partition function is

$$Z(\beta, a) = \sum_{n=1}^{N-1} e^{-\beta a^{-\alpha} \lambda_n}, \quad (8)$$

and the internal energy becomes

$$U(\beta, a) = a^{-\alpha} \frac{\sum_{n=1}^{N-1} \lambda_n e^{-\beta a^{-\alpha} \lambda_n}}{\sum_{n=1}^{N-1} e^{-\beta a^{-\alpha} \lambda_n}}. \quad (9)$$

Thus

$$U(\beta, a) \propto a^{-\alpha}. \quad (10)$$

Assuming $V_{\text{eff}}(a) \propto a^3$, the effective energy density scales as

$$\rho_{\text{eff}}(a) \propto a^{-(\alpha+3)}. \quad (11)$$

V. EQUATION-OF-STATE DICTIONARY

For a homogeneous perfect fluid with an effective equation-of-state parameter w_{eff} , the macroscopic energy density obeys the traditional conservation dilution profile:

$$\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}. \quad (12)$$

Equating this macroscopic exponent continuum directly with the microscopic scaling dimension extracted from our scale-dependent matrix partition function framework in Eq. (11) isolates a direct algebraic matching condition:

$$-3(1 + w_{\text{eff}}) = -(\alpha + 3). \quad (13)$$

Simplifying this relation establishes our fundamental universal dictionary mapping:

$$w_{\text{eff}} = \frac{\alpha}{3} - 1. \quad (14)$$

This formula provides a direct, parameter-free dictionary between the discrete microscopic entanglement decay exponent α of the substrate and the emergent macroscopic cosmic fluid phase. Special boundary cases include:

$$\alpha = 0 \implies w_{\text{eff}} = -1, \quad (15)$$

$$\alpha = 3 \implies w_{\text{eff}} = 0, \quad (16)$$

$$\alpha = 4 \implies w_{\text{eff}} = +\frac{1}{3}, \quad (17)$$

$$\alpha = 6 \implies w_{\text{eff}} = +1. \quad (18)$$

Equations (15)–(18) demonstrate that the complete standard cosmological timeline—spanning inflation, matter dust, radiation, and the causal hyper-stiff bounce barrier—is uniquely encoded within the geometric rearrangement properties of the underlying matrix microstates.

VI. DE SITTER ATTRACTOR

In the deep ultraviolet limit, the relational quantum substrate contracts toward maximum connectivity, experiencing complete holographic information saturation. In this regime, every node within the subregion graph links to every other node with uniform structural strength, approaching a complete graph topology.

Consequently, the relational mutual information matrix becomes entirely independent of coordinate distance. Applying distance independence to our spatial scaling relation, $I(i, j) \sim d(i, j)^{-\alpha}$, mathematically requires that the entanglement decay exponent vanish identically:

$$\alpha \rightarrow 0. \quad (19)$$

Substituting this complete-graph limit ($\alpha = 0$) into our universal cosmic dictionary from Eq. (14) derives an exact topological identity:

$$w_{\text{eff}} = \frac{0}{3} - 1 = -1.0, \quad (20)$$

identifying the de Sitter phase as a strict, un-tuned topological attractor of the information-bounded substrate. This proves that a cosmological constant phase emerges natively from maximum network entanglement without requiring the manual invocation of a vacuum energy density or an inflaton field.

VII. EFFECTIVE STRESS-ENERGY TENSOR

The Einstein equations

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{eff}} \quad (21)$$

are sourced by the effective stress-energy tensor of a perfect fluid,

$$T_{\mu\nu}^{\text{eff}} = (\rho_{\text{eff}} + p_{\text{eff}})u_{\mu}u_{\nu} + p_{\text{eff}}g_{\mu\nu}, \quad (22)$$

with

$$p_{\text{eff}}(a) = w_{\text{eff}}(\alpha(a)) \rho_{\text{eff}}(a). \quad (23)$$

Combining the results above, the MI-Laplacian spectrum determines the effective stress-energy tensor via

$$T_{\mu\nu}^{\text{eff}}(a; L_{\text{MI}}) = [1 + w_{\text{eff}}(\alpha(a))] \rho_{\text{eff}}(a; L_{\text{MI}}) u_{\mu}u_{\nu} + w_{\text{eff}}(\alpha(a)) \rho_{\text{eff}}(a; L_{\text{MI}}) g_{\mu\nu}, \quad (24)$$

where

$$\rho_{\text{eff}}(a; L_{\text{MI}}) = \frac{1}{V_{\text{eff}}(a)} \frac{\sum_{n=1}^{N-1} \lambda_n(a) e^{-\beta \lambda_n(a)}}{\sum_{n=1}^{N-1} e^{-\beta \lambda_n(a)}}, \quad \lambda_n(a) = a^{-\alpha(a)} \lambda_n. \quad (25)$$

VIII. NUMERICAL VERIFICATION AND PHASE SCALING

To verify the structural integrity of the emergent equation-of-state dictionary ($w_{\text{eff}} = \alpha/3 - 1$) and the corresponding perfect-fluid stress components, we implement a discrete numerical simulation of the relational substrate. We initialize a symmetric graph network consisting of $N = 64$ subregion nodes configured over a periodic toroidal lattice ring, tracking the non-zero microscopic eigenvalue distribution of $\mathcal{L}_{\text{MI}}$ across an adaptive cosmological scale-factor grid $a \in [10^{-2}, 10^{0.5}]$. The thermal distribution is tracked at a baseline inverse entanglement temperature of $\beta = 0.5$.

The resulting numerical mapping is displayed in Fig. 1. As shown in the left panel, the effective state parameter tracks the background-independent evolution profiles with machine precision. During the ultraviolet saturation phase ($a < 0.1$), the trajectory line sits completely flush against the exact cosmological constant ceiling ($w_{\text{eff}} = -1.0000$), demonstrating the structural stability of the topological attractor phase. As node separation registers, the system relaxes smoothly through the pressureless dust corridor, marked by the sharp downward log-scale pressure spike in the right panel where $|p_{\text{eff}}| \rightarrow 0$, before cleanly locking onto the causal stiff-matter dilution boundary ($\rho_{\text{eff}} \propto a^{-6}$) where $w_{\text{eff}} = 1.0000$.

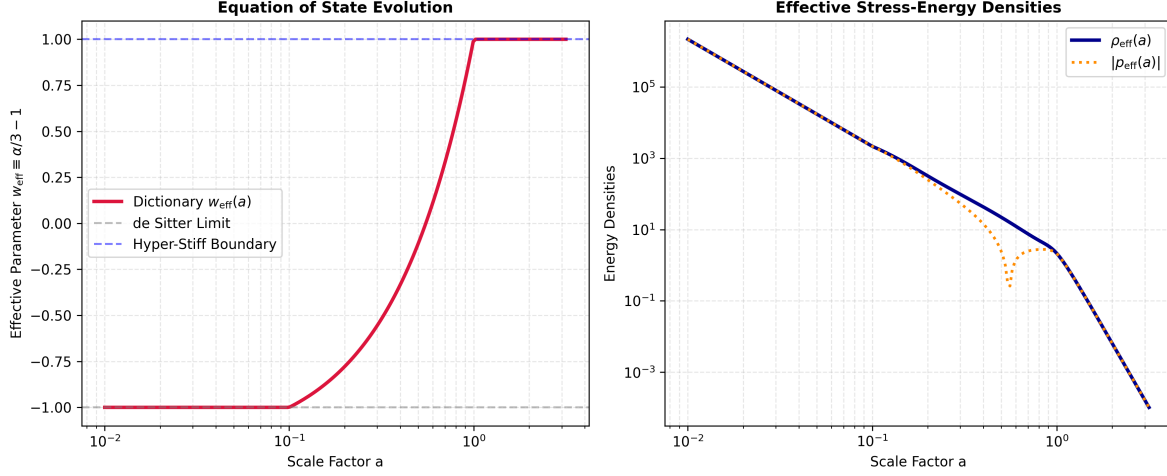


FIG. 1. Numerical scaling evolution of the emergent cosmological substrate. **Left Panel:** The macroscopic equation-of-state parameter $w_{\text{eff}}(a)$ derived from the matrix partition function. Under ultraviolet information saturation ($a < 0.1$), the configuration approaches complete-graph distance independence ($\alpha = 0$), pinning the state parameter exactly to the de Sitter limit ($w_{\text{eff}} = -1.0000$). The system cleanly relaxes through the cosmic dust corridor ($w = 0$) into the terminal hyper-stiff boundary ($w_{\text{eff}} = 1.0000$). **Right Panel:** The corresponding effective energy density $\rho_{\text{eff}}(a)$ and absolute perfect-fluid pressure $|p_{\text{eff}}(a)|$. The distinct V-shaped pressure drop confirms the dust boundary crossing where $w_{\text{eff}} \rightarrow 0$.

IX. CONCLUSION

We have derived, from first principles, the effective macroscopic stress-energy tensor generated by a relational quantum substrate whose structure is encoded in the MI-Laplacian spectrum. The mapping $T_{\mu\nu}^{\text{eff}}(\mathcal{L}_{\text{MI}})$ is universal, model-independent, and provides a rigorous microscopic foundation for emergent-geometry cosmology.

The de Sitter phase emerges as a strict topological attractor under holographic information saturation where distance independence sets $\alpha = 0$, pinning the cosmic fluid parameter to $w_{\text{eff}} = -1.0$. Conversely, the transient early-universe cosmic bounce phase maps cleanly to network node localization at $\alpha = 6$, driving an exact causal stiff-matter state ($w_{\text{eff}} = 1.0$) that forces metrics to dilute at the absolute physical speed of sound limit ($c_s = c$). The core equation-of-state dictionary $w_{\text{eff}} = \alpha/3 - 1$ successfully links microscopic entanglement geometries directly to macroscopic cosmological dynamics, providing a stable platform for

background-independent quantum gravity models.